Business Intelligence – Exercise Sheet: K-Means, k-NN, and Support Vector Machines for Wirtschaftsinformatik

Bachelor in Wirtschaftsinformatik, 6th Semester

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Exercises on K-Means Clustering, k-Nearest Neighbors, and Support Vector Machines

Exercises 1 to 7 concern K-means clustering for customer, process, and ERP data. Exercises 8 to 14 concern k-nearest-neighbor classification and regression. Exercises 15 to 20 concern linear support vector machines, margins, hinge loss, kernels, and BI interpretation.
Use exact expressions whenever convenient and round numerical answers to three decimals when needed.
For K-means, use the Euclidean distance

$$d(x, z) = \sqrt{\sum_{j=1}^{p} (x_j - z_j)^2}.$$

For a cluster C_k, the centroid is

$$\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i.$$

The K-means objective, also called within-cluster sum of squares or inertia, is

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2.$$

For k-nearest neighbors, classify a new observation by majority vote among the k closest training observations. In distance-weighted k-NN, use weights

$$w_i = \frac{1}{d_i + \varepsilon},$$

where d_i is the distance from the new point to neighbor i, and $\varepsilon > 0$ avoids division by zero.
For a linear support vector machine, use the decision function

$$f(x) = w^\top x + b.$$

The predicted class is

$$\hat{y} = \operatorname{sign}(f(x)),$$

with labels $y \in \{-1, +1\}$. The geometric distance of a point x to the separating hyperplane is

$$\frac{|w^\top x + b|}{\|w\|}.$$

The functional margin of an observation (x_i, y_i) is

$$m_i = y_i(w^\top x_i + b).$$

The hinge loss is

$$L_i = \max\{0, 1 - y_i(w^\top x_i + b)\}.$$

For a soft-margin SVM, one common objective is

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} L_i.$$

1. **Ex 1 (20 Points):** A BI analyst wants to cluster small business customers according to

   $$x_1 = \text{monthly software spend in hundred EUR}, \qquad x_2 = \text{number of support tickets}.$$

   The four customers are:

   $$A = (1,1), \qquad B = (2,1), \qquad C = (6,5), \qquad D = (7,5).$$

   The initial K-means centroids are

   $$\mu_1^{(0)} = (1,1), \qquad \mu_2^{(0)} = (7,5).$$

   (a) Compute the Euclidean distance from each customer to both initial centroids. (8 Points)

   (b) Assign each customer to the nearest centroid. (4 Points)

   (c) Compute the updated centroids $\mu_1^{(1)}$ and $\mu_2^{(1)}$. (5 Points)

   (d) Interpret the two clusters in CRM language. (3 Points)

2. **Ex 2 (20 Points):** An ERP system stores process observations with two variables:

   $$x_1 = \text{processing time in minutes}, \qquad x_2 = \text{number of manual corrections}.$$

   The observations are:

   $$P_1 = (2,1), \quad P_2 = (3,1), \quad P_3 = (4,2), \quad P_4 = (9,6), \quad P_5 = (10,7), \quad P_6 = (11,7).$$

   After one K-means iteration, the cluster assignment is

   $$C_1 = \{P_1, P_2, P_3\}, \qquad C_2 = \{P_4, P_5, P_6\}.$$

(a) Compute the centroid of C_1. (4 Points)

(b) Compute the centroid of C_2. (4 Points)

(c) Compute the within-cluster sum of squares J. (8 Points)

(d) Explain what a high value of J would mean in a process-mining dashboard. (4 Points)

3. **Ex 3 (20 Points):** A digital-commerce team clusters customers by

$$x_1 = \text{visits per week}, \qquad x_2 = \text{orders per month}.$$

The data are:

$$A = (1, 0), \quad B = (2, 0), \quad C = (3, 1), \quad D = (8, 5), \quad E = (9, 6), \quad F = (10, 6).$$

Two possible clusterings are proposed.
Clustering I:

$$C_1 = \{A, B, C\}, \qquad C_2 = \{D, E, F\}.$$

Clustering II:

$$C_1 = \{A, B\}, \qquad C_2 = \{C, D, E, F\}.$$

(a) Compute the centroids for Clustering I. (4 Points)

(b) Compute the K-means objective J for Clustering I. (5 Points)

(c) Compute the centroids for Clustering II. (4 Points)

(d) Compute the K-means objective J for Clustering II. (5 Points)

(e) Which clustering is preferred by K-means? Justify numerically. (2 Points)

4. **Ex 4 (20 Points):** A BI team clusters products by

$$x_1 = \text{profit margin in percent}, \qquad x_2 = \text{return rate in percent}.$$

The current centroids are

$$\mu_1 = (20, 5), \qquad \mu_2 = (40, 15).$$

A new product has feature vector

$$x = (30, 8).$$

(a) Compute the Euclidean distance from x to μ_1. (5 Points)

(b) Compute the Euclidean distance from x to μ_2. (5 Points)

(c) Assign the product to the nearest cluster. (4 Points)

(d) Explain how the assignment could be used in product-portfolio management. (3 Points)

(e) State one limitation of using only profit margin and return rate for product clustering. (3 Points)

5. **Ex 5 (20 Points):** A logistics dashboard clusters suppliers using

$$x_1 = \text{average delivery delay in days}, \qquad x_2 = \text{defect rate in percent}.$$

The supplier observations are:

$$S_1 = (1,1), \quad S_2 = (2,1), \quad S_3 = (1,2), \quad S_4 = (8,7), \quad S_5 = (9,8).$$

The initial centroids are

$$\mu_1^{(0)} = (1,1), \qquad \mu_2^{(0)} = (8,7).$$

(a) Perform the assignment step for all suppliers. (8 Points)

(b) Compute the updated centroids. (5 Points)

(c) Compute the K-means objective J after the update. (5 Points)

(d) Name the operational meaning of the two resulting supplier groups. (2 Points)

6. **Ex 6 (20 Points):** A bank wants to cluster customers by

$$x_1 = \text{account balance in thousand EUR}, \qquad x_2 = \text{number of digital logins per month}.$$

The raw observations are:

$$A = (1,2), \quad B = (2,3), \quad C = (50,4), \quad D = (60,5).$$

(a) Explain why K-means on the raw variables may be dominated by account balance. (4 Points)

(b) Compute the min-max normalized values of both variables using

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}.$$

(8 Points)

(c) Using the normalized data, discuss whether C and D are still necessarily close to A and B in terms of digital behavior. (4 Points)

(d) Explain why scaling is important before applying distance-based BI methods. (4 Points)

7. **Ex 7 (20 Points):** A company evaluates K-means cluster quality using the silhouette coefficient. For one customer i, the average distance to all other points in its own cluster is

$$a(i) = 2.0.$$

The average distance from this customer to the nearest other cluster is

$$b(i) = 5.0.$$

The silhouette value is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

   (a) Compute $s(i)$. (5 Points)

   (b) Interpret the value in terms of cluster separation. (4 Points)

   (c) Compute $s(i)$ again for $a(i) = 4.5$ and $b(i) = 5.0$. (5 Points)

   (d) Explain why the second case is less convincing from a BI-segmentation perspective. (3 Points)

   (e) State one reason why a high silhouette value does not automatically imply business usefulness. (3 Points)

8. **Ex 8 (20 Points):** A k-NN classifier predicts whether a customer will buy an add-on service. The training data are:

$$x_1 = \text{number of previous purchases}, \qquad x_2 = \text{website visits last week}.$$

Customer	x_1	x_2	Bought add-on?
A	1	1	no
B	2	1	no
C	4	3	yes
D	5	4	yes
E	6	5	yes

A new customer has

$$x_{\text{new}} = (3, 2).$$

   (a) Compute the Euclidean distance from x_{new} to each training customer. (8 Points)

   (b) Identify the $k = 3$ nearest neighbors. (4 Points)

   (c) Predict the class using majority vote. (4 Points)

   (d) Explain the prediction in sales-dashboard language. (4 Points)

9. **Ex 9 (20 Points):** A helpdesk system uses k-NN to classify whether an IT ticket will be escalated. The variables are:

$$x_1 = \text{waiting time in hours}, \qquad x_2 = \text{number of previous similar incidents}.$$

Ticket	x_1	x_2	Escalated?
1	1	0	no
2	2	1	no
3	3	1	no
4	5	3	yes
5	6	4	yes
6	7	5	yes

A new ticket has

$$x_{\text{new}} = (4, 2).$$

(a) Compute all Euclidean distances. (8 Points)

(b) Predict the class using $k = 1$. (3 Points)

(c) Predict the class using $k = 3$. (4 Points)

(d) Explain why $k = 1$ may be more sensitive to noise than $k = 3$. (5 Points)

10. **Ex 10 (20 Points):** A BI analyst applies distance-weighted k-NN for lead conversion. The three nearest neighbors of a new lead are:

Neighbor	Distance d_i	Class	Weight $w_i = 1/d_i$
1	1.0	converted	
2	2.0	not converted	
3	4.0	converted	

Use $\varepsilon = 0$ because all distances are positive.

(a) Compute the weight of each neighbor. (6 Points)

(b) Compute the total weighted vote for class converted. (4 Points)

(c) Compute the total weighted vote for class not converted. (4 Points)

(d) Predict the class. (3 Points)

(e) Explain why distance weighting may be useful in CRM applications. (3 Points)

11. **Ex 11 (20 Points):** A k-NN regression model predicts the monthly cloud cost of a customer. For a new customer, the four nearest neighbors have the following cloud costs in EUR:

$$y_1 = 100, \qquad y_2 = 120, \qquad y_3 = 180, \qquad y_4 = 200.$$

(a) Compute the k-NN regression prediction for $k = 2$, using the two nearest neighbors y_1, y_2. (4 Points)

(b) Compute the prediction for $k = 4$. (4 Points)

(c) Suppose the distances to the four neighbors are $1, 2, 3, 4$. Compute the distance-weighted prediction using weights $w_i = 1/d_i$. (8 Points)

(d) Explain why k-NN regression predictions may change strongly when k is changed. (4 Points)

12. **Ex 12 (20 Points):** A retailer classifies whether a customer is likely to return an online order. The variables are:

$$x_1 = \text{basket value in EUR}, \qquad x_2 = \text{delivery delay in days}.$$

The new observation is

$$x_{\text{new}} = (100, 3).$$

The nearest-neighbor search on unscaled data gives large influence to basket value.

(a) Explain why basket value may dominate Euclidean distance. (4 Points)

(b) Given basket values ranging from 20 to 220, normalize $x_1 = 100$ using min-max scaling. (4 Points)

(c) Given delivery delays ranging from 0 to 10, normalize $x_2 = 3$ using min-max scaling. (4 Points)

(d) Explain how the nearest neighbors may change after scaling. (4 Points)

(e) State one practical BI rule for deciding when scaling is required. (4 Points)

13. **Ex 13 (20 Points):** A k-NN classifier is evaluated on a test set for predicting whether invoices are paid late. The confusion matrix is:

	Predicted late	Predicted on time
Actual late	16	8
Actual on time	6	30

Treat late as the positive class.

(a) Identify TP, FP, TN, and FN. (4 Points)

(b) Compute accuracy. (4 Points)

(c) Compute precision. (4 Points)

(d) Compute recall. (4 Points)

(e) Explain whether the classifier is more problematic for false alarms or missed late payments. (4 Points)

14. **Ex 14 (20 Points):** A company compares k-NN models for customer churn with different values of k. The validation results are:

Model	k	Validation accuracy	Comment
A	1	0.86	very flexible
B	5	0.89	moderate smoothing
C	25	0.81	very smooth

   (a) Which model has the highest validation accuracy? (3 Points)

   (b) Explain why $k = 1$ can overfit the training data. (4 Points)

   (c) Explain why $k = 25$ can underfit local customer patterns. (4 Points)

   (d) Which model would you select for deployment based on the table? Justify. (4 Points)

   (e) State two additional checks before deploying the selected k-NN model in a churn dashboard. (5 Points)

15. **Ex 15 (20 Points):** A linear SVM is trained to separate two customer classes:

$$+1 = \text{premium customer}, \qquad -1 = \text{standard customer}.$$

The model is

$$f(x) = w^\top x + b, \qquad w = (2, 1), \qquad b = -5.$$

For the customers

$$A = (1, 1), \quad B = (2, 2), \quad C = (3, 1), \quad D = (4, 2),$$

   (a) Compute $f(x)$ for each customer. (8 Points)

   (b) Assign the predicted class to each customer using $\text{sign}(f(x))$. (4 Points)

   (c) Write the equation of the separating line. (3 Points)

   (d) Compute $\|w\|$. (3 Points)

   (e) Explain what the sign of $f(x)$ means in BI classification language. (2 Points)

16. **Ex 16 (20 Points):** An SVM for risky-invoice detection uses

$$w = (1, 2), \qquad b = -6.$$

The hyperplane is

$$w^\top x + b = 0.$$

   (a) Write the equation of the separating hyperplane in terms of x_1 and x_2. (3 Points)

   (b) Compute the geometric distance of $x = (2, 1)$ to the hyperplane. (5 Points)

(c) Compute the geometric distance of $x = (4, 2)$ to the hyperplane. (5 Points)

(d) Determine on which side of the hyperplane each point lies. (4 Points)

(e) Explain why points close to the hyperplane are operationally uncertain. (3 Points)

17. **Ex 17 (20 Points):** A linear SVM is trained with labels $y \in \{-1, +1\}$. The model is

$$w = (1, 1), \qquad b = -3.$$

The training observations are:

Observation	x_1	x_2	y
A	1	1	-1
B	2	1	-1
C	3	2	+1
D	4	3	+1

(a) Compute $f(x) = w^\top x + b$ for each observation. (6 Points)

(b) Compute the functional margin $m_i = y_i f(x_i)$ for each observation. (6 Points)

(c) Identify observations with $m_i = 1$, if any. (3 Points)

(d) Explain why such observations are called support vectors in the hard-margin case. (3 Points)

(e) State whether all observations are correctly classified. (2 Points)

18. **Ex 18 (20 Points):** A soft-margin SVM for churn prediction has

$$w = (1, -1), \qquad b = 0.$$

Three training observations are:

Customer	x_1	x_2	y
A	3	1	+1
B	1	3	-1
C	2	2	+1

Use the hinge loss

$$L_i = \max\{0, 1 - y_i(w^\top x_i + b)\}.$$

(a) Compute $f(x)$ for all three customers. (6 Points)

(b) Compute the hinge loss of each customer. (6 Points)

(c) Compute the total hinge loss. (3 Points)

(d) Compute $\frac{1}{2}\|w\|^2$. (3 Points)

(e) Explain why customer C is problematic for the SVM. (2 Points)

19. **Ex 19 (20 Points):** A BI team compares two linear SVM classifiers for a two-dimensional customer dataset.

Model A:

$$w_A = (2, 0), \qquad b_A = -4.$$

Model B:

$$w_B = (1, 1), \qquad b_B = -4.$$

(a) Compute $\|w_A\|$. (4 Points)

(b) Compute $\|w_B\|$. (4 Points)

(c) Compute the margin width $2/\|w\|$ for both models. (6 Points)

(d) Which model has the larger geometric margin? (3 Points)

(e) Explain why SVMs prefer larger margins, assuming similar training errors. (3 Points)

20. **Ex 20 (20 Points):** A company wants to classify customers as

$$+1 = \text{high churn risk}, \qquad -1 = \text{low churn risk}.$$

The data are not linearly separable in the original variables:

$$x_1 = \text{contract age}, \qquad x_2 = \text{monthly usage}.$$

The BI team considers a polynomial kernel

$$K(x, z) = (x^\top z + 1)^2.$$

Two customer vectors are

$$x = (1, 2), \qquad z = (3, 1).$$

(a) Compute the ordinary dot product $x^\top z$. (4 Points)

(b) Compute the kernel value $K(x, z)$. (4 Points)

(c) Explain, without deriving the full feature map, why a polynomial kernel can represent nonlinear decision boundaries. (5 Points)

(d) State one advantage of using a kernel SVM instead of manually constructing many interaction variables. (3 Points)

(e) State two BI risks of deploying a complex kernel SVM in a management dashboard. (4 Points)